FP16 Floating-Point Adder - Updated Block Diagram

DENORM-enabled architecture from FP16_Adder_Group_4_2305166_2305169_fixed.circ

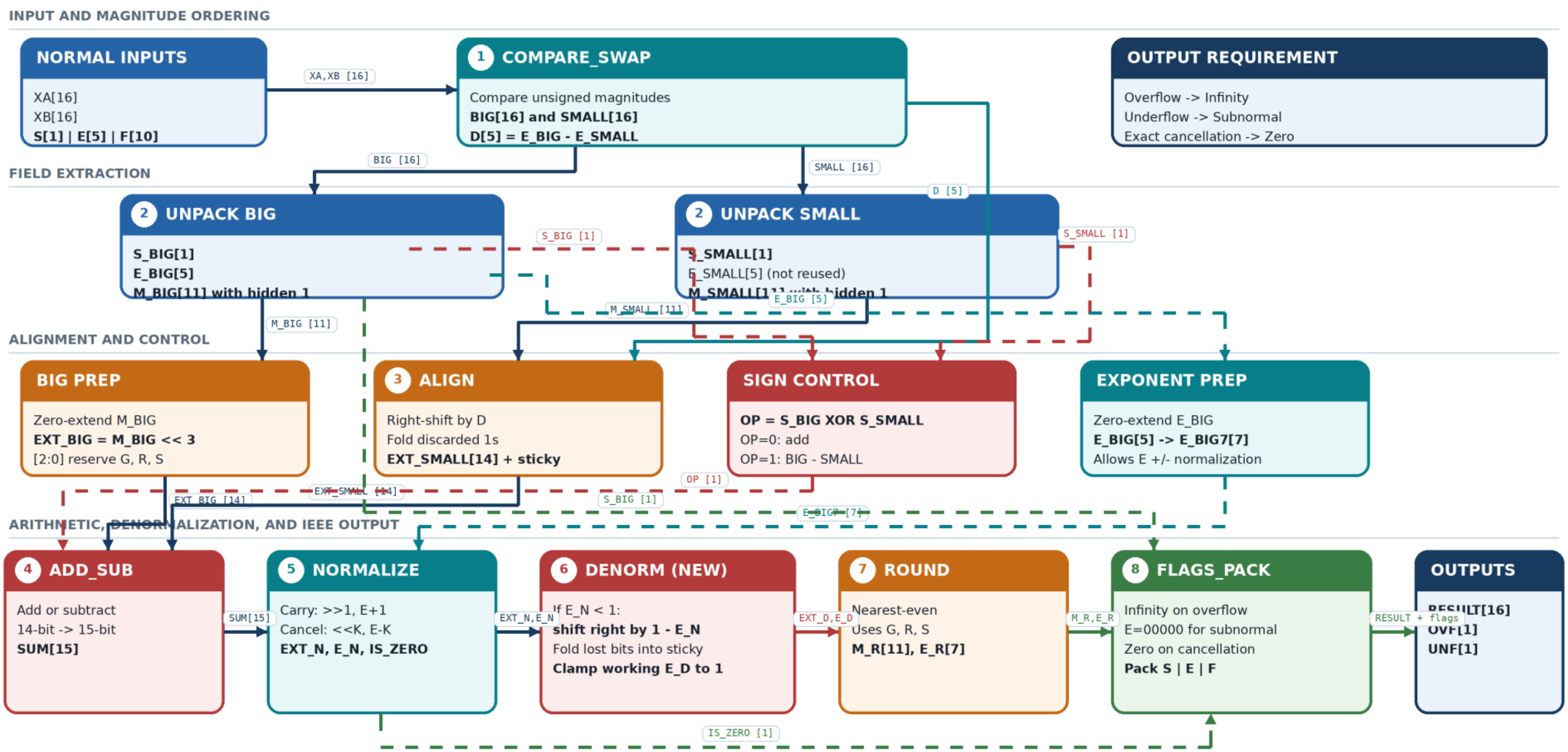

Figure 1. Updated top-level data path; DENORM is inserted between NORMALIZE and ROUND.

Input assumption: normal FP16 values only **Output handling:** Infinity / Subnormal / Zero in IEEE FP16 format **EXT bus:** significand[13:3] | G[2] | R[1] | S[0]

Module Interfaces and Special-Output Contract

Signal widths and behavior reproduced from the updated Logisim hierarchy.

#	Module	Inputs	Core action	Outputs
1	**COMPARE_SWAP**	XA[16], XB[16]	Order the operands by unsigned magnitude and compute the stored-exponent difference.	BIG[16], SMALL[16], D[5]
2	**UNPACK x2**	BIG[16] / SMALL[16]	Split sign, exponent, and fraction; restore the hidden leading 1 because inputs are normal.	S[1], E[5], M[11]
3	**ALIGN**	M_SMALL[11], D[5]	Append guard/round/sticky positions, right-shift the smaller significand, and preserve discarded 1s in sticky.	EXT_SMALL[14]
4	**ADD_SUB**	OP, EXT_BIG[14], EXT_SMALL[14]	OP=0 adds; OP=1 subtracts SMALL from BIG. A 15th bit preserves carry.	SUM[15]
5	**NORMALIZE**	SUM[15], E_BIG7[7]	Shift for carry or cancellation, update the exponent, and detect exact cancellation.	EXT_N[14], E_N[7], IS_ZERO
6	**DENORM**	EXT_N[14], E_N[7]	For E_N<1, shift right by 1-E_N, merge discarded bits into sticky, and clamp the working exponent to 1; otherwise pass through.	EXT_D[14], E_D[7]
7	**ROUND**	EXT_D[14], E_D[7]	Round to nearest-even with guard, round, sticky, and retained LSB; allow a rounding carry into the exponent.	M_R[11], E_R[7]
8	**FLAGS_PACK**	S_BIG, E_R[7], M_R[11], IS_ZERO	Select normal/subnormal/infinity/zero encoding, generate flags, and pack the IEEE FP16 fields.	RESULT[16], OVF, UNF

IEEE FP16 output cases

Result class	Condition used by the output path	FP16 bit representation	Flags
Normal finite	0 < E_R < 31 and M_R[10]=1	S_BIG | E_R[4:0] | M_R[9:0]	OVF=0, UNF=0
Overflow	E_R >= 31 and not exact zero	S_BIG | 11111 | 0000000000 (Infinity)	OVF=1
Underflow	M_R[10]=0, finite, and not exact zero	S_BIG | 00000 | M_R[9:0] (Subnormal)	UNF=1
Exact cancellation	IS_ZERO=1	0000000000000000 (+0)	OVF=0, UNF=0

Why E_D becomes 1: ROUND may promote a subnormal to the smallest normal; FLAGS_PACK then decides exponent 00000 from M_R[10]. **Exact zero:** IS_ZERO overrides the packed sign and fields, producing 0x0000.